READING ROOM

Pneumonia from chest X-rays

We test whether focusing on the lungs improves normal-versus-pneumonia classification.

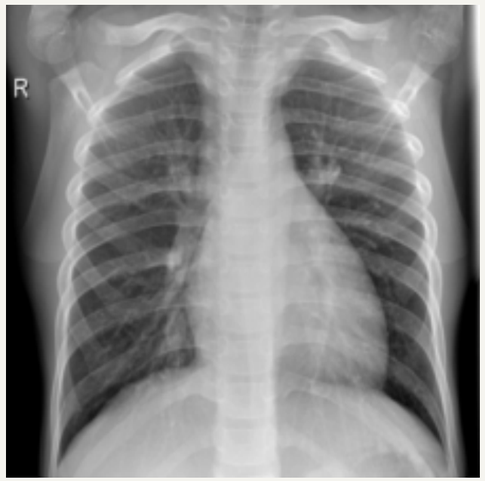


Samarth Shekhar Meghavi Rao

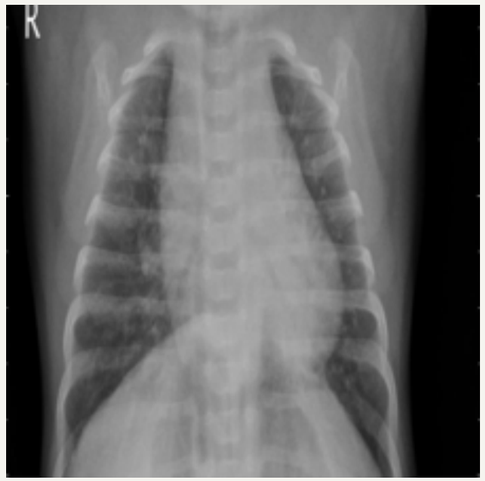
Rachit Kumar Harshal Nerpagar

Computer Vision / M1-M5

Dataset and prediction task



Normal



Pneumonia

Public Kermany dataset (2018)

Pediatric chest X-rays with two labels: Normal and Pneumonia.

5,856 original image records

Final split: 4,164 training / 1,042 validation / 624 test records.

Test: 234 Normal + 390 Pneumonia

Source: Kermany et al., Guangzhou Women and Children's Medical Center.

Original split: 5,216 training / 16 validation / 624 test records.

One pediatric source center; the benchmark is not an external clinical evaluation.

Our journey: M1 to M5

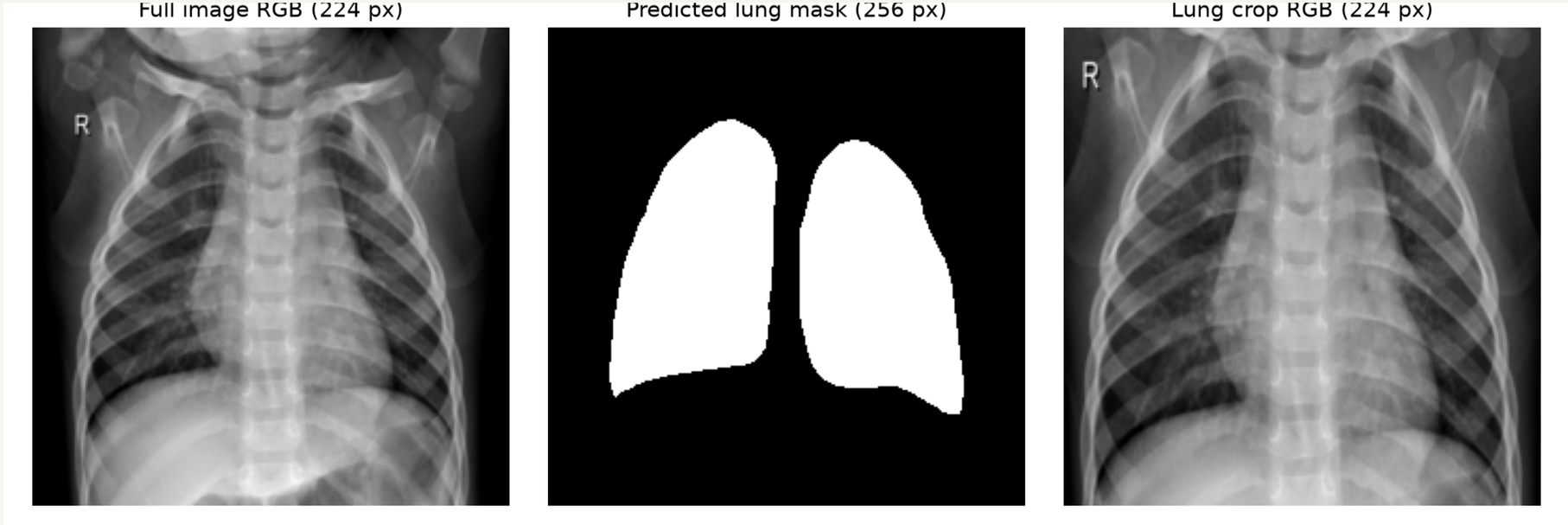
We kept the failed attempts and audited them before improving the system.

Stage	What we did	Result
M1	Literature review and initial lung-focused idea	Proposal
M2	First cached CNN training (5 epochs)	87.18%
M2	Longer cached audit (15 epochs)	85.74%
M2	Corrected online processing + fresh augmentation	92.15%
M3	Recorded crop + transformer hypothesis	Prediction
M4	Four matched comparisons; final crop CNN-ViT	**93.75%**
M5	Errors, verification, report and working demo	Verified

Historical runs changed several settings. Only A/B/C/D provide the matched comparison of cropping and the extra model.

From the full X-ray to a lung crop

Crop the original pixels first; resize only after the lung-focused region is chosen.



1 / Full image

Start with the complete X-ray, including lungs and surrounding anatomy.

2 / Predicted mask

Fixed PSPNet estimates both lungs. White means predicted lung pixels, not disease.

3 / Lung crop RGB

Keep a rectangle plus 10% margin. Resize to 224 x 224 in three-channel RGB format.

ResNet18 + DeiT-Tiny classify the crop; final D combines their scores with equal weights.

Four matched comparisons

Same split and training; vary the input region and whether the transformer is added.

	Full image	Lung crop
CNN	**A** **86.86%** Full X-ray, CNN only: the matched control.	**B** **92.63%** Lung crop, same CNN: B versus A tests cropping.
CNN + ViT	**C** **88.14%** Full X-ray plus transformer: C versus A tests the extra model.	**D** **93.75%** Crop plus both models: D versus B tests ViT on crops.

CNN = ResNet18; ViT = DeiT-Tiny. Four heads trained five epochs; encoders stayed fixed.

Final model results

Condition D: lung crop + CNN/ViT, evaluated on 624 benchmark test records.

93.75%

Accuracy

585 / 624 images correct

98.21%

Pneumonia recall

383 / 390 pneumonia cases found

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32 false alarms

Normal-labelled images predicted as pneumonia.

7 missed cases

Pneumonia-labelled images predicted as normal.

Against the matched full-image CNN, accuracy increased by 6.89 percentage points.

Mistakes, limits and next steps

These are actual wrong predictions; a good benchmark score does not remove errors.



False alarm

A normal-labelled image was predicted as pneumonia.



Missed pneumonia

A pneumonia-labelled image was predicted as normal.

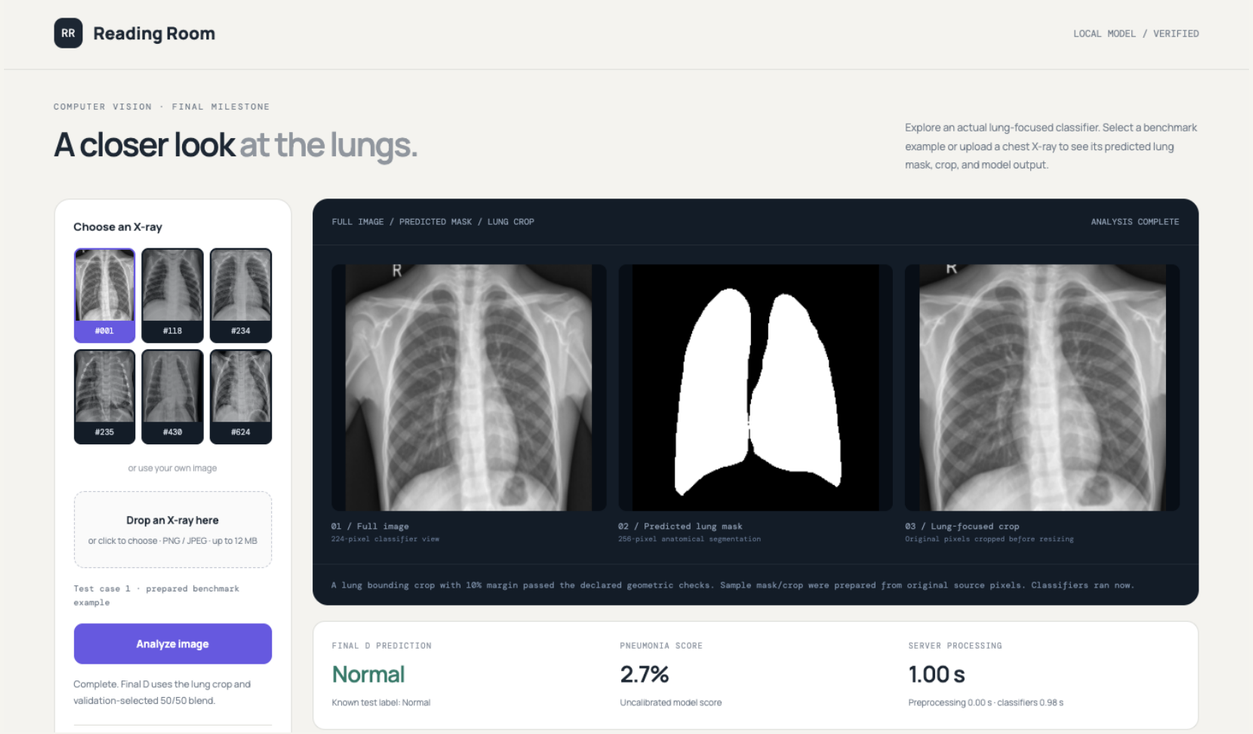
Limit: cropping helped test results but reduced performance on this validation split.

Next: repeat group-aware splits, test independent data and evaluate limited fine-tuning.

Poor masks use a full-image fallback (82 / 5,824 unique images). Masks do not mark disease.

Live demonstration and this term's work

We delivered a measured crop/ViT comparison, verified model, web app and GitHub package.



1. Choose an image

2. Click Analyze

3. Show mask + result

Built-in masks are prepared; classifiers run now. Uploads run segmentation too.

Backup: open the recorded offline demo and explain that it contains saved actual model responses.

Slides, guide, code and model

hulkintherapy.github.io/reading-room-pneumonia/